

HYDROGENATED NBR SYNTHESIS AND PROCESS LANDSCAPE FROM SELECTED PATENTS

1. ABSTRACT

This report synthesizes eight patents relevant to hydrogenated nitrile rubber (HNBR), with emphasis on how hydrogenation level, molecular weight, catalyst choice, and downstream processing are controlled to tailor performance. The target material is defined as nitrile rubber derived from conjugated diene/ α,β -unsaturated nitrile copolymers that have been fully or partially hydrogenated to improve heat resistance, oil resistance, ozone/weather resistance, and processability. Across the selected patents, two major synthesis directions emerge: conventional selective hydrogenation of NBR in solution or latex using noble-metal catalysts, and metathesis-assisted hydrogenation/depolymerization using ruthenium/osmium catalysts to reduce molecular weight and narrow polydispersity in a single step. A third theme is not primary synthesis but formulation/process integration, where HNBR is compounded or bonded into high-performance articles such as silicone blends or PVDF-lined flexible pipes.

The patents on low-molecular-weight HNBR (US7951875B2, US7772328B2, US9150669B2) focus on simultaneous hydrogenation and metathesis in the presence of hydrogen and co-olefins, using ruthenium/osmium catalyst systems related to Grubbs/Hoveyda/Arlt/Wilkinson-type chemistry. These disclosures emphasize lower Mw, narrower MWD, and reduced Mooney viscosity relative to conventional HNBR feedstocks. EP2473280B1 similarly uses simultaneous hydrogenation and metathesis, but specifically compares catalyst families and shows that catalyst identity strongly affects hydrogenation rate and residual double bond content. In contrast, EP3827029A1 addresses conventional HNBR hydrogenation with ruthenium, palladium, or rhodium compounds and focuses on minimizing catalyst residues while preserving ageing properties; it highlights the tradeoff between catalyst loading, reaction time, and post-removal complexity. US20220073666A1 shifts to latex hydrogenation of PEG acrylate-functionalized NBR, followed by coagulation, to produce processable functional HNBR terpolymers. Finally, US20130303700A1 and US10641418B2 demonstrate how HNBR is used in blends and bonded pipe structures, with process windows centered on peroxide curing, scorch time, adhesion, and thermal stability rather than polymer formation itself.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20130303700A1
- Patent Title: Blends of partially hydrogenated nitrile rubber and silicone rubber, ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses rubber blends made by mixing partly hydrogenated nitrile rubber with vinyl-containing silicone rubber in the presence of a metathesis catalyst, followed by kneading/mixing and subsequent vulcanization. The process is aimed at improving compatibility and blend morphology rather than synthesizing HNBR itself. - HNBR hydrogenation extent: at least 50% up to 99% — Claim 3 - HNBR hydrogenation extent: 75 to 98.5% — Claim 3 - HNBR hydrogenation extent: 80 to 98% — Claim 3 - HNBR hydrogenation extent: 85 to 96% — Claim 3 - Organic solvent amount: not more than 1200 ppm — Claim 2 - Organic solvent amount: not more than 500 ppm — Claim 2 - Organic solvent amount: not more than 100 ppm — Claim 2 - Therban 3467 acrylonitrile content: 34% by weight — Example 1 - Therban 3467 residual double bond content: 5.5% — Example 1 - Therban 3467 Mooney viscosity: 68 ME — Example 1 - Siloprene VR vinyl group content: 5 mol % — Example 1 - Siloprene VR Mn: 138000 g/mol — Example 1 - Siloprene VR Mw: 304000 g/mol — Example 1 - Brabender capacity: 350 ml — Example 1 - Fill level: 70% — Example 1 - Rotor speed: 50 min⁻¹ — Example 1 - Start temperature: 35° C. — Example 1 - Initial kneading time: 2 min — Example 1 - Catalyst addition window: within 5 min — Example 1 - Internal temperature: 60° C. — Example 1 - Kneading time: 60 min — Example 1 - Room-temperature storage between mixing stages: 24 h — Example 1

Relevant Experimental Evidence - Example 1: HNBR/silicone blends were prepared with and without metathesis catalyst, showing the role of catalyst addition in blend formation. - Example 1: Multi-stage mixing at 60° C. with 24 h storage between stages was used to build the inventive blends. - Example 1: SEM was used to assess phase morphology of unvulcanized blends containing 10 wt% or 20 wt% VMQ.

Technical Relevance This patent is relevant because it uses partially hydrogenated NBR as the HNBR component and provides practical compounding conditions for HNBR-based elastomer systems. It shows how residual unsaturation and catalyst-enabled processing can be leveraged in downstream HNBR formulations.

Patent 2

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a single-step process in which nitrile rubber is subjected to simultaneous hydrogenation and metathesis in the presence of hydrogen and optionally a co-olefin, using a ruthenium/osmium catalyst system. The goal is to produce low-molecular-weight HNBR with narrower molecular weight distribution and improved processability. - Mw range: 30000 to 250000 g/mol — Claim 1 - Mooney viscosity: less than 10 ML 1+4 @100° C. — Claim 1 - MWD/PDI: less than 2.5 — Claim 1 - Mw range: 40000 to 220000 g/mol — Claim 2 - PDI: less than 2.3 — Claim 3 - Co-olefin pressure: 500 psi — Table 1 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Cement concentration: 15 wt.-% solid — Example 1 - Cement concentration: 6 wt.-% solid — Example 3 - Rubber mass: 200 g — Examples 1-2 - MCB mass: 1133 g — Examples 1-2 - Rubber mass: 75 g — Examples 3-4 - MCB mass: 1175 g — Examples 3-4 - Degassing cycles: 3 — Examples 1-4 - Co-olefin: C₂H₄ — Examples 1-3 - Co-olefin: 1-hexene — Example 4 - 1-hexene amount: 150 g — Example 4 - Mn: 27 kg/mol — Table 2 - Mw: 54 kg/mol — Table 2 - PDI: 2.0 — Table 2 - Mooney viscosity: 3 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene was used as co-olefin at 500 psi and 80° C. with 0.05 phr catalyst, giving reduced Mw and Mooney viscosity. - Example 2: Increasing catalyst load to 0.10 phr with ethylene still produced low-MW HNBR but with different molecular-weight profile. - Example 3: Lower cement concentration and 0.25 phr catalyst gave the lowest Mooney viscosity and narrowest PDI in the table. - Example 4: 1-hexene was used instead of ethylene, demonstrating that co-olefin identity affects product properties.

Technical Relevance This patent is directly relevant because it targets the core synthesis problem for HNBR: reducing molecular weight while hydrogenating the nitrile rubber backbone. It provides a practical catalyst/process window for making low-viscosity HNBR suitable for improved processing.

Patent 3

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses the same general low-molecular-weight HNBR concept as US7951875B2: simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen and optionally a co-olefin, using ruthenium/osmium catalyst systems. It emphasizes controlled depolymerization/hydrogenation to obtain HNBR with reduced Mw and narrower MWD. - Mw range: 30000 to 250000 g/mol — Claim 1 - Mooney viscosity: less than 10 ML 1+4 @100° C. — Claim 1 - MWD/PDI: less than 2.5 — Claim 1 - Mw range: 40000 to 220000 g/mol — Claim 2 - PDI: less than 2.3 — Claim 3 - Co-olefin pressure: 500 psi — Table 1 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Cement concentration: 15 wt.-% solid — Example 1 - Cement concentration: 6 wt.-% solid — Example 3 - Rubber mass: 200 g — Examples 1-2 - MCB mass: 1133 g — Examples 1-2 - Rubber mass: 75 g — Examples 3-4 - MCB mass: 1175 g — Examples 3-4 - Degassing cycles: 3 — Examples 1-4 - Co-olefin: C₂H₄ — Examples 1-3 - Co-olefin: 1-hexene — Example 4 - 1-hexene amount: 150 g — Example 4 - Mn: 27 kg/mol — Table 2 - Mw: 54 kg/mol — Table 2 - PDI: 2.0 — Table 2 - Mooney viscosity: 3 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene co-olefin and 0.05 phr catalyst at 80° C. produced a lower-MW HNBR product. - Example 2: Higher catalyst loading with ethylene was evaluated for its effect on molecular-weight reduction. - Example 3: The lowest Mooney viscosity in the set was obtained under the 6 wt% cement condition. - Example 4: 1-hexene co-olefin was shown to be a viable alternative to ethylene.

Technical Relevance This patent is relevant because it discloses a controlled route to low-molecular-weight HNBR, which is a key target for improved processability. The examples provide concrete process variables that influence hydrogenation/metathesis efficiency and final polymer rheology.

Patent 4

- Patent Number: EP2473280B1
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen, optionally a co-olefin, and a ruthenium/osmium catalyst. It is aimed at producing HNBR with low residual double bond

content, reduced molecular weight, and narrower molecular weight distribution in a single step. - RDB at 0 min: 100% — Table 2 - RDB at 60 min: 2.0% — Table 2, HNBR 1 - RDB at 60 min: 0.7% — Table 2, HNBR 2 - RDB at 60 min: 15.4% — Table 2, HNBR 3 - RDB at 60 min: 2.9% — Table 2, HNBR 4 - RDB at 120 min: 0.4% — Table 2, HNBR 1 - RDB at 120 min: 0.5% — Table 2, HNBR 2 - RDB at 120 min: 3.1% — Table 2, HNBR 3 - RDB at 120 min: 0.6% — Table 2, HNBR 4 - RDB at 180 min: 0.3% — Table 2, HNBR 1 - RDB at 180 min: 0.3% — Table 2, HNBR 2 - RDB at 180 min: 1.3% — Table 2, HNBR 3 - RDB at 180 min: 0.3% — Table 2, HNBR 4 - RDB at 240 min: 0.6% — Table 2, HNBR 3 - Catalyst amount: 0.057 phr — Table 1, HNBR 1 - Catalyst amount: 0.080 phr — Table 1, HNBR 2 - Catalyst amount: 0.06 phr — Table 1, HNBR 3 - Catalyst amount: 0.055 phr — Table 1, HNBR 4 - Metal content: 65.6 ppm — Table 1, HNBR 1 - Metal content: 91.3 ppm — Table 1, HNBR 2 - Metal content: 65.6 ppm — Table 1, HNBR 3 - Metal content: 65.6 ppm — Table 1, HNBR 4 - Mw: 191541 g/mol — Table 3, HNBR 1 - Mn: 68643 g/mol — Table 3, HNBR 1 - PDI: 2.79 — Table 3, HNBR 1 - Mooney viscosity: 38.2 — Table 3, HNBR 1 - Mw: 135772 g/mol — Table 3, HNBR 2 - Mn: 58259 g/mol — Table 3, HNBR 2 - PDI: 2.33 — Table 3, HNBR 2 - Mooney viscosity: 20.2 — Table 3, HNBR 2 - Mw: 195304 g/mol — Table 3, HNBR 3 - Mn: 64130 g/mol — Table 3, HNBR 3 - PDI: 3.05 — Table 3, HNBR 3 - Mooney viscosity: 69.3 — Table 3, HNBR 3 - Mw: 144212 g/mol — Table 3, HNBR 4 - Mn: 59851 g/mol — Table 3, HNBR 4 - PDI: 2.45 — Table 3, HNBR 4 - Mooney viscosity: 32.6 — Table 3, HNBR 4

Relevant Experimental Evidence - Example 1: Four catalyst systems were compared for hydrogenation rate and final polymer properties. - Example 1: Grubbs III showed faster hydrogenation than Grubbs II or Wilkinson's catalyst at equal metal content. - Example 1: HNBR 2 gave the lowest Mooney viscosity among the compared samples. - Example 1: Residual double bond content dropped from 100% to below 1% for the best-performing systems within 120-180 min.

Technical Relevance This patent is highly relevant because it directly compares catalyst systems for simultaneous hydrogenation/metathesis of NBR and quantifies the impact on RDB, Mw, and Mooney viscosity. It provides evidence for selecting catalyst families and loading to optimize HNBR synthesis.

Patent 5

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for producing low-molecular-weight HNBR by contacting nitrile rubber with a ruthenium/osmium catalyst in the presence of hydrogen and optionally a co-olefin, thereby combining hydrogenation and metathesis. The process is designed to give HNBR with low residual double bond content, reduced molecular weight, and narrower molecular weight distribution. - Mw range: 30000 to 250000 g/mol — Claim 1 - Mooney viscosity: less than 10 ML 1+4 @100° C. — Claim 1 - MWD/PDI: less than 2.5 — Claim 1 - Mw range: 40000 to 220000 g/mol — Claim 2 - PDI: less than 2.3 — Claim 3 - Catalyst amount: 0.041 phr — Table 1, HNBR 1 - Catalyst amount: 0.045 phr — Table 1, HNBR 2 - Catalyst amount: 0.060 phr — Table 1, HNBR 3 - Metal content: 65.6 ppm — Table 1, HNBR 1 - Metal content: 65.6 ppm — Table 1, HNBR 2 - Metal content: 65.6 ppm — Table 1, HNBR 3 - Co-catalyst: NIL — Table 1, HNBR 1 - Co-catalyst: NIL — Table 1, HNBR 2 - Co-catalyst: Triphenylphosphine — Table 1, HNBR 3 - Co-catalyst amount: 1.0 phr — Table 1, HNBR 3 - RDB at 0 min: 100% — Table 2 - RDB at 60 min: 2.5% — Table 2, HNBR 1 - RDB at 60 min: 7.0% — Table 2, HNBR 2 - RDB at 60 min: 15.4% — Table 2, HNBR 3 - RDB at 120 min: 0.6% — Table 2, HNBR 1 - RDB at 120 min: 4.5% — Table 2, HNBR 2 - RDB at 120 min: 3.1% — Table 2, HNBR 3 - RDB at 180 min: 0.5% — Table 2, HNBR 1 - RDB at 180 min: 3.9% — Table 2, HNBR 2 - RDB at 180 min: 1.3% — Table 2, HNBR 3 - RDB at 240 min: 0.4% — Table 2, HNBR 1 - RDB at 240 min: 3.5% — Table 2, HNBR 2 - RDB at 240 min: 0.7% — Table 2, HNBR 3 - Mw: 113371 g/mol — Table 3, HNBR 1 - Mn: 48352 g/mol — Table 3, HNBR 1 - PDI: 2.34 — Table 3, HNBR 1 - Mooney viscosity: 33.3 — Table 3, HNBR 1 - Mw: 133758 g/mol — Table 3, HNBR 2 - Mn: 54595 g/mol — Table 3, HNBR 2 - PDI: 2.45 — Table 3, HNBR 2 - Mooney viscosity: 41.3 — Table 3, HNBR 2 - Mw: 195304 g/mol — Table 3, HNBR 3 - Mn: 64130 g/mol — Table 3, HNBR 3 - PDI: 3.05 — Table 3, HNBR 3 - Mooney viscosity: 69.3 — Table 3, HNBR 3

Relevant Experimental Evidence - Example 1: Hoveyda, Arlt, and Wilkinson-type catalysts were compared at similar metal content. - Example 1: HNBR 1 achieved the fastest hydrogenation and lowest RDB among the three samples. - Example 1: The catalyst system influenced both molecular weight reduction and final Mooney viscosity. - Example 1: The industrial-norm catalyst (HNBR 3) gave the highest Mooney viscosity and highest PDI.

Technical Relevance This patent is relevant because it provides a catalyst-comparison framework for low-MW HNBR synthesis and shows how co-catalyst choice affects hydrogenation efficiency. It is useful for selecting catalyst systems that balance rapid hydrogenation with controlled depolymerization.

Patent 6

- Patent Number: US20220073666A1
- Patent Title: Process for producing peg acrylate-hnbr copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses latex hydrogenation of PEG acrylate-functionalized NBR followed by coagulation to produce PEG acrylate-HNBR copolymer. The process preserves functional PEG acrylate units while selectively hydrogenating the diene-derived unsaturation, yielding a processable HNBR terpolymer/functional copolymer. - PEG acrylate-NBR nitrile unit content: 10% to 60% by weight — Claim 1 - PEG acrylate-NBR diene unit content: 20% to 80% by weight — Claim 1 - PEG acrylate-NBR PEG acrylate unit content: 10% to 60% by weight — Claim 1 - Nitrile unit content: 15% to 40% by weight — Claim 2 - Diene unit content: 30% to 65% by weight — Claim 3 - PEG acrylate unit content: 10% to 50% by weight — Claim 4 - Nitrile unit content: 15% to 35% by weight — Claim 5 - Diene unit content: 35% to 60% by weight — Claim 5 - PEG acrylate unit content: 20% to 50% by weight — Claim 5 - PEG acrylate formula n: 1 to 8 — Claim 1 - PEG acrylate formula R1: hydrogen or methyl — Claim 1 - PEG acrylate formula R: hydrogen or branched or unbranched C1-C20-alkyl — Claim 1 - PEG acrylate preferred R: C1-C10-alkyl — Description - PEG acrylate preferred R: C1-C4-alkyl — Description - PEG acrylate preferred R: methyl, ethyl or butyl — Description - PEG acrylate preferred n: 2 to 8 — Description - PEG acrylate preferred n: 2 to 5 — Description - PEG acrylate preferred n: 2 or 3 — Description - Reaction temperature: 80° C. to 160° C. — Description - Hydrogen pressure: 3000000 to 8000000 pascals — Description - Catalyst class: transition metal catalyst from group 8, 9 or 10 — Description

Relevant Experimental Evidence - No numbered examples were detected by the parser.

Technical Relevance This patent is relevant because it extends HNBR synthesis to functionalized PEG acrylate-containing NBR feedstocks and uses latex hydrogenation, a major industrial route. It provides composition windows and process conditions for making processable functional HNBR materials.

Patent 7

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenation of NBR in solution using ruthenium, palladium, or rhodium compounds, with emphasis on minimizing catalyst residues and preserving ageing properties of the resulting HNBR. The process is framed around catalyst loading, hydrogenation conditions, and downstream catalyst recovery/removal rather than a novel polymerization route. - HNBR conjugated diene content: 40% to 90% by weight

— Claim 1 - HNBR α,β -unsaturated nitrile content: 10% to 60% by weight — Claim 1 - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200°C — Claim 1 - Hydrogenation pressure: 700000 to 1500000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Rhodium content in recovered HNBR: 58-60 ppm — Description - Rhodium content in purified HNBR: 5.8 to 9.2 ppm — Description - Catalyst amount: 0.45 parts — Description (EP-A-1454924 cited) - Ageing test: 4 days at 140°C — Description - Gel content increase: less than 10% — Description - Mooney viscosity increase: less than 40% — Description

Relevant Experimental Evidence - No numbered examples were detected by the parser.

Technical Relevance This patent is relevant because it addresses practical HNBR hydrogenation using noble-metal catalysts while controlling catalyst residue and ageing performance. It is especially useful for defining process windows and catalyst loadings for industrial HNBR production.

Patent 8

- Patent Number: US10641418B2
- Patent Title: ... gas-resistant, high-pressure, large-bore bonded flexible rubber pipe and method ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a bonded flexible pipe in which an HNBR-based peroxide-vulcanized rubber compound is directly bonded to a PVDF liner. The HNBR is not synthesized here, but the patent is relevant because it specifies the HNBR compound properties and curing conditions needed for high-performance HNBR-based articles. - Pipe internal diameter: 4" (102 mm) — Example 1 - Operating pressure: 69 MPa — Example 1 - PVDF liner thickness: 6 mm — Example 1 - HNBR compound layer thickness: 1 mm — Example 1 - Mooney 05, 150° C. scorch time: 12.0 minutes — Example 1 - Peroxide loading: 8 phr — Example 1 - Peroxide active agent: 2,5-dimethyl-2,5-di(t-butylperoxy)-3-hexyne — Example 1 - Peroxide active agent content: 45% — Example 1 - Peroxide half-life at 154° C.: 1 hour — Example 1 - Acrylonitrile content: 30% to 40% — Example 1 - Residual double bond content: below 2% — Example 1 - Carbon black loading: 70 phr — Example 1 - Triallyl isocyanate coagent:

2.5 phr — Example 1 - Other additives: 18.7 phr — Example 1 - Vulcanization temperature: 175° C. — Example 1 - Pipe internal diameter: 4" (102 mm) — Example 2 - Mooney 05, 150° C. scorch time: 5.6 minutes — Example 2 - Peroxide loading: 8 mass fractions — Example 2 - Vulcanization temperature: 155° C. — Example 2 - Adhesion value: 3.1 N/mm — Example 2 - Pipe internal diameter: 3" (76 mm) — Example 3 - Operating pressure: 103.5 MPa — Example 3 - HNBR compound layer thickness: 1 mm — Example 3 - Mooney 05, 150° C. scorch time: 14.0 minutes — Example 3 - Peroxide loading: 7 phr — Example 3 - Vulcanization temperature: 160° C. — Example 3 - Adhesion value: above 15 N/mm — Example 3 - PVDF copolymer: at least 80 mole % vinylidene-fluoride units — Claim 1 - PVDF liner melting point: above 150° C. — Claim 2 - Peroxide half-life at 150° C.: at least 1 hour — Claim 3 - Scorch-protected peroxide: yes — Claim 5 - Vulcanization temperature: at least 160° C. — Claim 1/Description

Relevant Experimental Evidence - Example 1: A 4-inch pipe with a 6 mm PVDF liner and HNBR compound cured at 175° C. achieved cohesive failure and >10 N/mm adhesion. - Example 2: A lower scorch-time compound and lower vulcanization temperature produced adhesive failure and only 3.1 N/mm adhesion. - Example 3: A 3-inch pipe with a 14-minute scorch time compound and 160° C. vulcanization achieved cohesive failure and >15 N/mm adhesion.

Technical Relevance This patent is relevant because it shows the downstream performance requirements for HNBR compounds in aggressive-service flexible pipes. It identifies scorch time, peroxide selection, and vulcanization temperature as key parameters for successful HNBR-based bonding and durability.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7951875B2, US7772328B2, US9150669B2, and EP2473280B1 all use simultaneous hydrogenation/metathesis concepts, but EP2473280B1 explicitly compares catalyst families and hydrogenation profiles, while the US patents emphasize low-MW product targets and co-olefin effects.
- US7951875B2 and US7772328B2 disclose essentially the same process window: 80° C., 500 psi co-olefin pressure, and catalyst loadings of 0.05-0.25 phr, with ethylene or 1-hexene as co-olefins.
- US9150669B2 differs by focusing on catalyst-system optimization and residue management; it uses 0.041-0.060 phr catalyst and adds a triphenylphosphine co-catalyst in one comparative case.
- EP2473280B1 reports catalyst amounts of 0.055-0.080 phr and metal contents of 65.6-91.3 ppm, with Grubbs III showing faster hydrogenation than Grubbs II or Wilkinson's catalyst at equal metal content.
- US20220073666A1 is a latex-hydrogenation route rather than a metathesis route, and it uses broader composition windows for PEG acrylate-functionalized NBR (10-60 wt% nitrile, 20-80 wt% diene, 10-60 wt% PEG acrylate).

- EP3827029A1 uses conventional solution hydrogenation with Ru/Pd/Rh catalysts at 60-200°C and 0.7-15 MPa, emphasizing catalyst residue control rather than molecular-weight reduction by metathesis.
- US20130303700A1 and US10641418B2 are downstream application patents: the former uses partly hydrogenated NBR in silicone blends with metathesis catalyst, while the latter uses HNBR in peroxide-vulcanized PVDF-bonded flexible pipes.
- Across the set, lower residual unsaturation is associated with better service performance, but processability is improved either by lowering Mw/Mooney via metathesis-assisted hydrogenation or by tailoring compound scorch time and cure conditions.

4. CONCLUSION

The patent set shows two dominant HNBR synthesis strategies: conventional selective hydrogenation of NBR in solution or latex, and simultaneous hydrogenation/metathesis to reduce molecular weight and Mooney viscosity in one step. For low-viscosity HNBR, the strongest evidence supports ruthenium/osmium metathesis-capable catalysts, co-olefin-assisted processing, and reaction temperatures around 80° C. with catalyst loadings in the 0.04-0.25 phr range. For conventional hydrogenation, the evidence supports 60-200°C, 0.7-15 MPa hydrogen pressure, and careful catalyst-residue management to preserve ageing properties. Functional HNBR latex routes are also viable, especially for PEG acrylate-containing copolymers. Overall, the best-supported synthesis targets are low residual double bond content, Mw in the 20,000-250,000 g/mol range, and controlled Mooney viscosity, with downstream cure/bonding performance improved by maintaining adequate scorch time and using peroxide systems matched to the application.

5. REFERENCES

- US20130303700A1 | Blends of partially hydrogenated nitrile rubber and silicone rubber, ... | Not disclosed | US | 2013 | <https://patents.google.com/patent/US20130303700A1/en>
- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- EP2473280B1 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473280B1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- US20220073666A1 | Process for producing peg acrylate-hnbr copolymer | Not disclosed | US | 2022 | <https://patents.google.com/patent/US20220073666A1/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>

- US10641418B2 | ... gas-resistant, high-pressure, large-bore bonded flexible rubber pipe and method ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10641418B2/en>